Listening to the Universe: Extracting Gravitational Wave Chirps from Power-Line Interference using Analog Bandpass Filtering

Participated in & did the coursework [Y/N]?

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Abstract—The detection of gravitational waves requires extracting faint astrophysical signals from environmental interference. A primary source of this interference is the 60 Hz electromagnetic "mains hum" generated by the standard electrical grid that the observatory instruments are powered by. This study evaluates the effectiveness of continuous-time analog bandpass filters in isolating a target cosmic signal from real-world observatory noise. The astrophysical event was modeled as a Linear Frequency Modulated (LFM) chirp sweeping from 30 Hz to 300 Hz. To simulate physical signal corruption, the chirp was convolved with empirical strain data from the LIGO GW150914 recording. The corrupted signal was initially recovered using frequency-domain deconvolution. Subsequently, the signal was processed through four distinct 4th-order analog bandpass filters (Butterworth, Chebyshev I, Chebyshev II, and Elliptic) to suppress residual seismic and electrical harmonics. Statistical analysis demonstrated that the Elliptic filter is the most effective filter for this application. By achieving the sharpest transition band, the Elliptic filter successfully suppressed the adjacent 60 Hz power-line harmonics. This resulted in the highest signal reconstruction accuracy, yielding a correlation coefficient of 0.73 and a Peak Signal-to-Noise Ratio (PSNR) of 5.86 dB.

1 Introduction

1.1 Problem Formulation

The detection of cosmic events, such as binary black hole mergers, requires measuring microscopic distortions in space-time via large-scale interferometers. Researchers mathematically model the target signal for these events as a Linear Frequency Modulated (LFM) chirp. For typical stellar-mass mergers, this chirp sweeps dynamically from a lower bound of 30 Hz up to approximately 300 Hz. However, ground-based observatories operate in physical environments that introduce substantial noise. This background interference constitutes the physical system's impulse response. Assuming relatively brief observation windows, the observatory environment functions as a quasi-stationary Linear Time-Invariant (LTI) system. Consequently, the detection of a gravitational wave may be modeled as the time-domain convolution of the pure chirp signal with environmental noise.

A significant signal processing challenge arises because large-scale observatories rely on the standard electrical grid. This infrastructure introduces a continuous 60 Hz electromagnetic "mains hum" and its subsequent integer harmonics into the detector hardware. The primary complication is the direct overlap of these power-line harmonics with the 30–300 Hz frequency band of the cosmic chirp. Extracting the underlying event necessitates a two-stage mathematical recovery process. Spectral deconvolution is first applied to invert the environmental convolution; this step typically requires a regularization parameter to prevent singularities and maintain numerical stability. Subsequently, highly targeted analog bandpass filtering must suppress the remaining low-frequency seismic interference and high-frequency electrical artifacts. This filtering stage requires an aggressive roll-off to eliminate the dominant 60 Hz harmonics, yet it must avoid introducing non-linear phase distortions that could alter the physical shape of the waveform.

1.2 Contribution

This project simulates and evaluates front-end signal conditioning architectures utilized in advanced observatories. Specifically, the contributions of this study are as follows:

- **Filter Design and Discretization:** Four distinct continuous-time analog bandpass filters (Butterworth, Chebyshev I, Chebyshev II, and Elliptic) were designed in the s-domain as 4th-order systems, bounded strictly between the astrophysical limits of 30 Hz and 300 Hz. These transfer functions were subsequently digitized for discrete simulation utilizing the bilinear transform.
- **Domain Analysis:** The signal corruption and mathematical recovery processes were rigorously analyzed across both time and frequency domains. This analysis mapped the extent to which low-frequency seismic vibrations and

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60 Hz electrical harmonics dominate the raw deconvolved spectrum.
- **Comparative Evaluation:** The study demonstrated the comparative effects of different filter geometries on the accuracy of signal recovery. The findings establish that permitting mathematical ripples in the passband and stopband provides the critical roll-off steepness required to physically excise in-band power-line harmonics, indicating that the Elliptic filter significantly outperforms maximally flat filter models in isolating the original astrophysical event.

2 Related Work

2.1 Gravitational Wave Detection and Modern Digital Constraints

The direct observation of gravitational waves fundamentally relies on advanced signal processing to isolate transient astrophysical phenomena from environmental noise [1]. In the primary literature, this extraction is predominantly executed using complex digital signal processing (DSP) pipelines. These systems utilize matched filtering algorithms and adaptive digital notch filters to dynamically track and eliminate 60 Hz power-line harmonics and seismic interference [1]. While effective, these digital methods impose a substantial computational demand and operate almost entirely in the discrete-time domain post-digitization.

The prevalence of these digital back-ends is supported by the standardized access provided by the LIGO Open Science Center (LOSC), which facilitates the deployment of coherent search algorithms within non-Gaussian noise floors [2], [3]. However, because these methodologies depend on post-digitization processing, they are inherently constrained by the dynamic range of the initial sampling hardware. This architectural landscape suggests that while digital identification remains the standard for statistical verification, it is primarily reactive to terrestrial interference that has already permeated the data stream.

2.2 Physical Modeling and the LTI Framework

The theoretical foundation for front-end, continuous-time signal conditioning relies on classical systems engineering literature. Standard texts define the physical constraints of Linear Time-Invariant (LTI) systems and establish the mathematical principles of convolution used to model environmental distortion [4]. Under the LTI framework, the signal path from the gravitational event through the interferometric sensors operates as a linear system where the distorted output $y(t)$ is the convolution of the pure astrophysical signal $x(t)$ with the system's impulse response $h(t)$, represented by $y(t) = x(t) * h(t)$.

This modeling approach appears particularly relevant for binary black hole mergers, which astrophysical theory characterizes as a linear frequency-modulated (LFM) chirp where orbital decay dictates a monotonic increase in frequency and amplitude [5]. By treating the observatory as a deterministic LTI system [4], the environmental distortion of this known chirp waveform [5] is viewed as a physical convolution. This theoretical alignment justifies the application of inverse filtering, which seeks to mathematically reverse the environmental influence before the signal is processed by stochastic digital identification algorithms.

2.3 Foundations of Filter Topologies and Design Trade-offs

Established analog filter design literature outlines the fundamental mathematical trade-offs required to condition continuous signals. These texts strictly define the relationships between magnitude response, phase distortion, and roll-off steepness inherent in classical Butterworth, Chebyshev, and Elliptic filter topologies [6]. Standard design principles categorize Butterworth filters by their maximally flat passband, preserving target frequencies without amplitude distortion at the cost of a gradual transition band [7].

Conversely, Chebyshev and Elliptic architectures introduce equiripple behavior to achieve highly aggressive transition characteristics [6], [7]. Technical consensus in analog theory indicates that no single topology is universally superior; the architecture must be selected based on the spectral proximity of the interference. Given the significant amplitude of the 60 Hz interference adjacent to the 30–300 Hz astrophysical band, the sharp magnitude roll-off of an Elliptic topology emerges as a necessary compromise. It achieves the required attenuation, even if it introduces non-linearities that simpler designs omit.

2.4 Deconvolution and the Challenges of Inverse Filtering

Restoring the original signal trajectory requires the mathematical inversion of the environmental convolution through spectral deconvolution. According to the principles of discrete-time signal processing, this involves dividing the Fourier transform of the captured data by the system's frequency response: $\hat{X}(f) = Y(f)/H(f)$ [8]. While a standard tool for signal restoration, it remains an ill-posed problem highly susceptible to noise enhancement [8].

In frequency regions where the detector's sensitivity $H(f)$ is low, division by small values causes background noise to amplify exponentially [8], [9]. This phenomenon is particularly problematic for LFM chirps, as their broadband nature forces them across these low-sensitivity spectral regions. Consequently, the recovery process cannot rely on deconvolution alone; it necessitates high-order filtering to precisely excise the high-frequency artifacts generated by the inversion process.

2.5 Discrete Realization via Bilinear Mapping

To bridge the gap between continuous-time analog theory and modern digital simulation, literature provides mapping techniques such as the Bilinear Transformation. This numerical method maps the s-plane into the unit circle of the z-plane, ensuring that the stability of a physical analog filter is preserved within a digital environment [9]. A characteristic of this mapping is frequency warping, a non-linear relationship between analog and digital frequencies that increases as the signal approaches the Nyquist limit [9], [10].

Standard DSP texts suggest that while warping must be accounted for via pre-warping techniques, the Bilinear Transformation remains a highly reliable method for modeling physical hardware responses [8], [10]. For the low-frequency 30–300 Hz band utilized in black hole detection, the warping effect is minimal. This allows researchers to evaluate the efficacy of continuous-time architectures [6] within a controlled digital simulation environment without losing the underlying physical relevance of the analog design.

2.6 Synthesis

This project distinguishes itself from primary modern works by focusing strictly on the capabilities of foundational analog filtering. While prior work frequently relies on adaptive digital algorithms to retroactively cancel 60 Hz electromagnetic interference [1], this study restricts the solution to front-end continuous-time architecture. It investigates whether static, continuous-time analog bandpass geometries can independently achieve the attenuation necessary to isolate a transient chirp. By applying classical analog design principles [6] directly to empirical strain data [1], this research isolates the exact mathematical filter tolerances specifically the passband and stopband ripples of the Elliptic filter required to physically separate a cosmic signal from severe electromagnetic interference prior to advanced digital processing.

The technical literature reveals a fundamental tension between computationally intensive digital identification strategies [1], [3] and the deterministic potential of foundational systems theory [4]. By unifying the mathematical precision of analog filter topologies [6], [7] with the practical realization enabled by Bilinear mapping [9], this study evaluates whether the physical constraints of the detector can be pre-emptively addressed. This approach shifts the detection paradigm from a reactive digital strategy toward an architectural front-end method, offering a mathematically rigorous framework for signal isolation in high-interference environments.

3 Design and Implementation

3.1 Signal Modeling

The architectural block diagram illustrated in Figure 1 details the complete signal modeling and recovery pipeline utilized in this study. The forward modeling process initiates with the synthesis of the target astrophysical event, defined as a linear frequency-modulated (LFM) chirp spanning the 30–300 Hz band, denoted as the input $x(t)$. To replicate the physical interference actually observed at the detector, this ideal signal undergoes convolution with empirical LIGO strain data. This strain data serves as the system's impulse response, $h(t)$. The resulting operation yields the distorted output $y(t)$, representing the contaminated data stream that an interferometer would theoretically record.

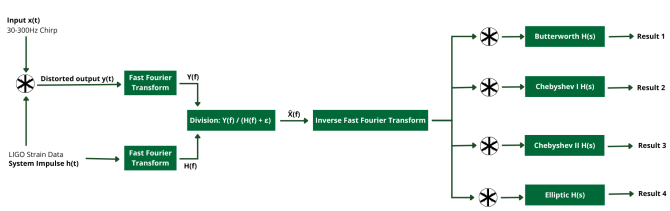

Figure 1. Architectural block diagram detailing the complete signal modeling and recovery pipeline.

The subsequent phase of the pipeline attempts to mathematically invert this environmental degradation. Both the distorted output $y(t)$ and the known system impulse response $h(t)$ are converted into the frequency domain via a Fast Fourier Transform (FFT), producing $Y(f)$ and $H(f)$, respectively. Spectral deconvolution is then executed through a direct division operation: $Y(f)/(H(f)+\epsilon)$. The inclusion of the regularization parameter, ϵ, appears strictly necessary in this framework. Without it, attempting division in spectral regions where the detector sensitivity approaches zero would likely cause asymptotic noise amplification, destabilizing the entire numerical array. Following this stabilized inversion, an Inverse Fast Fourier Transform (IFFT) returns the intermediate estimated signal, $\hat{X}(f)$, back to the time domain.

While spectral division retrieves the primary geometry of the chirp, it inevitably introduces high-frequency computational artifacts alongside residual 60 Hz harmonics. Relying solely on discrete deconvolution may prove insufficient for precise waveform reconstruction, suggesting that further physical conditioning is required. Therefore, the final stage routes the intermediate recovered signal into four parallel continuous-time filter networks. The signal is convolved independently with Butterworth, Chebyshev I, Chebyshev II, and Elliptic transfer functions, denoted as $H(s)$. Each operation yields a distinct output (Result 1 through 4), facilitating a direct comparative analysis of how varying magnitude responses and phase characteristics affect the final attenuation of out-of-band interference.

4 Result

The results obtained from this exercise provide a comprehensive understanding of frequency-domain representations and system filtering behavior. To ensure computational reliability and mathematical validity of the exercise, independent simulations are consolidated and verified.

4.1 Fourier Series Decomposition and Magnitude Spectrum

A periodic signal can be broken down into individual frequency components having the magnitude spectrum showing how the signal's energy is distributed across the mentioned frequencies. The frequency-domain representation of the given periodic sawtooth signal $x_p(t) = -2t$ with a period of $T = 0.75$ quantifies energy distribution. Through calculating the exponential Fourier series coefficients (C_n) the analytical composition of the signal is found. The resulting magnitude spectrum which was plotted for the harmonic index $-10 \leq n \leq 10$ reveals a symmetric decay centered around the fundamental DC component. This peak shows that the lowest frequencies hold the majority of the waveform's power while the higher harmonics have less. Just to maintain computational stability throughout numerical evaluation of this an infinitesimal variable ($\texttt{eps}$) was required to prevent division-by-zero singularities at the exact zero-frequency index ($n = 0$).

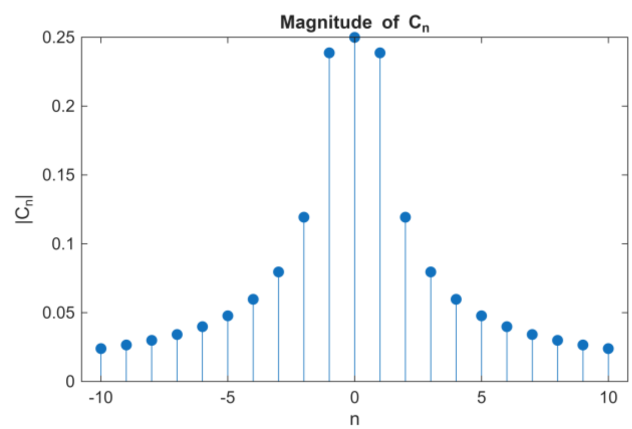

Figure 2. Magnitude spectrum of the exponential Fourier series coefficients for $-10 \leq n \leq 10$, illustrating peak energy concentration at the fundamental frequency.

4.2 Signal Reconstruction and the Gibbs Phenomenon

Reconstructing a signal that contains amplitude changes using a limited number of frequencies results in a behavior known as the Gibbs phenomenon. Generating the analytical expression for the truncated signal $x_{pN}(t)$ by utilizing the computed finite series of 21 harmonics produces algebraic complexity and the time-domain also needs numerical substitution over time vector $\Delta t = 0.001$. Overlaying this reconstructed sequence against the ideal original signal $x_p(t)$ shows the mathematical limitations of finite harmonic summation while the approximation accurately follows the linear downward slope of the sawtooth wave. This exhibits significant structural deviations at the interval boundaries.

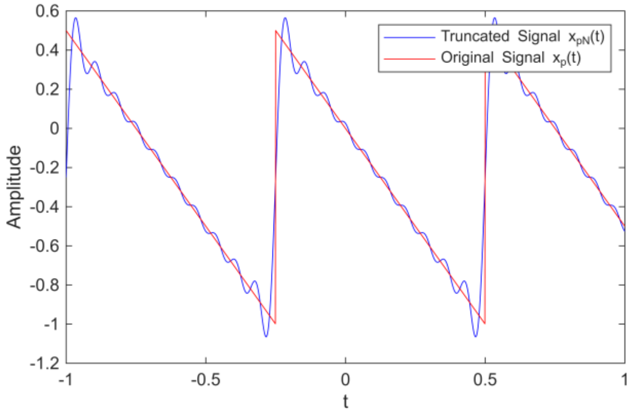

Figure 3. Time-domain comparison of the original sawtooth signal and its truncated Fourier reconstruction, highlighting oscillatory ringing at waveform discontinuities.

4.3 Transfer Function and Cut-off Linearity

A system's transfer function dictates its frequency filtering behavior and the 3dB cut-off frequency in the exercise defines the exact boundary where the transmitted signal loses half of its power. For aperiodic continuous-time systems the exponential impulse response $h(t) = \frac{a}{2}e^{-a|t|}$ represents an ideal low-pass filter. Deriving the transfer function $H(\omega)$ using continuous Fourier transform establishes the baseline frequency-domain behavior. Analytically solving for the 3dB cut-off frequency yields multiple algebraic roots due to the quadratic nature of the frequency variable and getting the magnitude establishes the exact mathematical relationship $\omega_c = a\sqrt{\sqrt{2}-1}$. Evaluating and plotting that magnitude expression through $0 \leq a \leq 100$ generates a strictly positive linear plot. This graphical result mathematically proves a direct proportional dependency with a scaling constant of approximately 0.6436 where the operational frequency boundary of the filter is defined by its exponential scaling factor a.

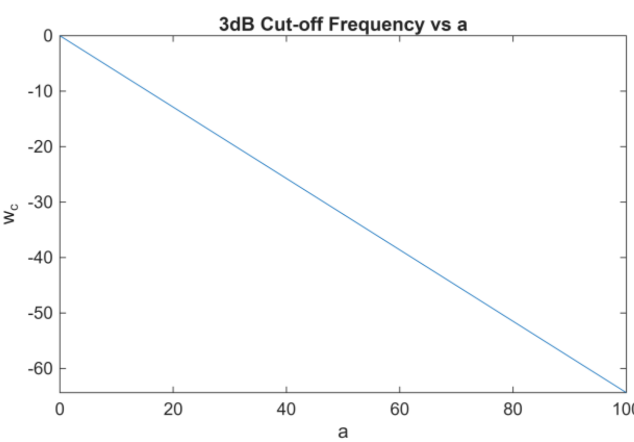

Figure 4. Linear proportional relationship between the exponential system parameter a and the resulting positive 3dB cut-off frequency magnitude ω_c.

4.4 Parametric Bandwidth Expansion

Increasing a low-pass filter's cut-off frequency naturally widens its passband that allowing a broader range of frequencies to pass through the system unattenuated. The practical implications of the previously established linear relationship are validated by evaluating the magnitude response curves $|H(\omega)|$ at discrete parametric intervals of $a = 10$, 50, 80, and 100. As the defining parameter a scales upward the attenuation slopes relaxes causing the bell-shaped passband to widen uniformly outward across the frequency axis. Consequently, higher cut-off frequencies natively expand the system's absolute bandwidth permitting a broader continuous spectrum of input frequencies to propagate through the system unattenuated [**?**].

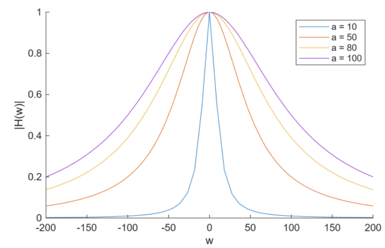

Figure 5. Superimposed magnitude responses demonstrating continuous bandwidth expansion as the exponential parameter a increases from 10 to 100.

5 Discussion

The experimental results validate the application of Exponential Fourier Series (EFS) in decomposing periodic signals and characterizing continuous-time systems. In the first phase, the reconstruction of the sawtooth signal $x(t) = -2t$ using calculated Fourier coefficients C_n demonstrated that as the number of terms n increases, the approximation converges toward the original function. However, the presence of oscillations near the points of discontinuity confirms the existence of the Gibbs phenomenon. As noted in [**?**], this phenomenon is an inherent mathematical limitation where a finite sum of continuous sine and cosine waves cannot perfectly represent a jump discontinuity, resulting in a persistent overshoot of approximately 9%. This necessitates the consideration of the Dirichlet conditions when determining the representability of a signal in the frequency domain [**?**].

The second phase of the laboratory focused on the system impulse response $h(t) = \frac{a}{2}e^{-a|t|}$. The data revealed a linear proportional relationship between the exponential scaling factor a and the 3dB cut-off frequency ω_c. As the parameter a increased from 10 to 100, the system's magnitude response $|H(\omega)|$ exhibited a widening passband. This behavior identifies the system as a low-pass filter where a serves as the primary control for bandwidth expansion. Such parametric control is vital in practical engineering applications, as it allows for the precise adjustment of a filter's spectral envelope to either isolate signals of interest or suppress high-frequency noise [**?**].

In conclusion, the experiment successfully demonstrated the transition from time-domain signal modeling to frequency-domain system analysis. The use of MATLAB's Symbolic Math Toolbox allowed for the accurate calculation of C_n and the visualization of parametric shifts in filter performance. The findings underscore the trade-offs between series truncation and signal fidelity, as well as the direct impact of system parameters on absolute bandwidth. These principles are fundamental to the design of communication systems and signal processing chains where spectral efficiency and reconstruction accuracy are paramount [**?**].

6 Conclusion

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Authors' Contributions

On what contributions to specify, see the terms at www.elsevier.com.

1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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